

BAYESIAN METHOD OF MOMENTS (BMOM) ANALYSIS OF PARAMETRIC AND SEMIPARAMETRIC REGRESSION MODELS

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Summary: The Bayesian Method of Moments is applied to standard regression models using alternative series expansions of an unknown regression function. We describe estimation loss functions, predictive loss functions and posterior odds as techniques to determine how many terms in a particular expansion to keep and how to choose among different types of expansions. the developed theory is then applied in a Monte-Carlo experiment to data generated from a CES production function.

1. Introduction

In this paper, we discuss how the Bayesian Method of Moments (BMOM) can be used to analyze parametric and semiparametric regression models. We discuss how to choose among alternative regression models when series expansions have been used to approximate an unknown regression function,

and how to determine the number of terms to keep in a particular series expansion.

The BMOM approach has been described and applied in previous work by Zellner (1996, 1997b, 1997c), Zellner and Sacks (1996), Green and Stawderman (1996), Tobias and Zellner (1999) and LaFrance (1999). In this paper we introduce several model selection criteria and also describe how the BMOM approach can be used. One of the distinguishing features of the BMOM approach is that it yields post-data densities for models' parameters without use of an assumed likelihood function. Since use of BMOM post-data densities for models' parameters yields a solution to the famous inverse problem put forward by Bayes (1763), the approach is called Bayesian. In addition, use of the BMOM yields a predictive density for as yet unobserved data, e.g. future values of the dependent variable in a regression. These BMOM predictive densities have been employed in Tobias and Zellner (1999) along with observed future values of the dependent and independent variables to compute posterior odds on alternative BMOM models and Bayesian models, the latter based on assumed likelihood functions and prior densities for parameters.

In the present paper, it will be shown how the BMOM approach can be employed to analyze semiparametric models. Here, we take up the case of analyzing the performance of alternative series expansions, which are known to converge in mean square to the unknown regression function. As regards earlier analyses of these semiparametric models, Ryu (1993) recognizes that the parameters of these models can be estimated and analyzed using least squares, Pearson's method of moments, maximum likelihood or Traditional Bayesian (henceforth just Bayesian) approaches. Clearly, maximum likelihood

and Bayesian approaches require that the form of the likelihood function be assumed, and thus are fully parametric. While the least-squares and method of moments approaches do not require use of a likelihood function, they just yield large sample or asymptotic inference results that may not be accurate in finite samples. Further, if post-data densities for parameters and for future observations are desired, least squares, method of moments, and maximum likelihood approaches will not produce them nor will a Bayesian approach without knowledge of the form of the likelihood function. Herein, we shall concentrate on producing BMOM finite sample estimates, parameter and predictive densities, tests, etc. that do not require use of an assumed likelihood function or of asymptotic approximations.

The plan of the paper is as follows. In section II, we present a canonical BMOM analysis of a regression model in which the assumptions of the approach and results yielded by it, point estimates, parameter and predictive densities, post-data odds, etc. will be derived and discussed. Then in section III, BMOM procedures for analyzing and comparing alternative series expansions for an unknown regression function are developed and described. Applications of these results using generated data are given in section IV. Section V contains a summary of results and some concluding remarks about future research.

2. BMOM analysis of a canonical regression

In this section, we assume that an investigator has chosen a particular series expansion (e.g. Fourier or Legendre expansion), with k terms to approximate a particular regression function, and that the model for the n given values of

the dependent variable, $y' = (y_1, y_2, \dots, y_n)$, is

$$y = X\beta + u \quad (1)$$

where β is a $k \times 1$ vector of regression parameters, including an intercept parameter, with unknown values and u is an $n \times 1$ vector of realized error terms. The $n \times k$ matrix X , assumed of rank k , has a column of ones as its first column and terms of the particular series expansion as elements of each row of X . Here we are assuming that the particular expansion and the number of terms retained, k , are considered to be an adequate representation of the underlying unknown regression function. See below where this assumption is appraised using data and post-data odds for alternative models.

2.1 Post-data moments and densities of parameters

To assume that the model in (1) is adequate implies, among other things, that there are no systematic elements in the realized error vector, u , correlated with variables in X . This assumption is formalized as assumption 1 in the BMOM approach as follows:

Assumption 1 $X'E(u|D) = 0$

where $E(u|D)$ denotes the post-data mean of u given the data. That is, the given, unknown values of the elements of the realized error vector u are considered subjectively random, just as in Bayesian analyses of realized error terms; see, for example, Chaloner and Brant (1988), Zellner and Moulton (1985) and Zellner (1975). Thus assumption 1 indicates that the columns of X are orthogonal to $E(u|D)$. Or from (1),

$$y = XE(\beta|D) + E(u|D) \quad (2)$$

and assumption 1 implies that the observation vector, y , is the sum of two orthogonal vectors. Note also since the first column of X is an $n \times 1$ vector of ones, denoted by q , we have from assumption 1, $q'E(u|D) = 0$, or $E\left(1/n \sum_{i=1}^n u_i\right) = E(\bar{u}|D) = 0$. Thus, given that we assume that we have an appropriate form and an adequate number of terms included in (1), the expectation of the mean of the realized error terms is assumed equal to zero, that is there is no systematic component in the realized error terms.

Further, if we write

$$(X'X)^{-1}X'y = \beta + (X'X)^{-1}X'u \quad (3)$$

and take the post-data expectation of both sides of (3), using assumption 1, we obtain,

$$E(\beta|D) = (X'X)^{-1}X'y = \hat{\beta}. \quad (4)$$

That is, the post-data expectation of the regression coefficient vector is equal to the least squares estimate. Also, it is known that a post-data mean is an optimal point estimate relative to a quadratic loss function in terms of post-data expected loss. Further, the post data mean of the realized error vector u in (1) is

$$E(u|D) = y - X\hat{\beta} = \hat{\mu}, \quad (5)$$

which is just the least squares residual vector. In addition, note that $u - \hat{\mu} = X(X'X)^{-1}X'(u - \hat{\mu})$, given that $X'\hat{\mu} = 0$. Then the post-data covariance matrix for u , denoted by $V(u|D)$, must satisfy the following functional equation:

$$V(u|D) \equiv E((u - \hat{\mu})(u - \hat{\mu})'|D) = X(X'X)^{-1}X'V(u|D)X(X'X)^{-1}X'. \quad (6)$$

Since there are only k free elements of u in the n equations in (1), $V(u|D)$ must be of rank k . Thus we introduce the following assumption that fixes

the form of the realized error covariance matrix up to a multiplicative positive scalar multiplier

Assumption 2 $V(u | \sigma^2, D) = X(X'X)^{-1}X'\sigma^2$,

where $\sigma^2 \equiv 1/n \sum_{i=1}^n u_i^2 = u'u/n$. It is seen that the parameter σ^2 is represented as an average of the sum of squared deviations of the realized errors from their expected mean of zero, $E(\bar{u} | D) = 0$ which follows from assumption 1. Note that the covariance matrix in assumption 2 satisfies the functional equation in (6). Given assumption 2 and $\beta - \hat{\beta} = -(X'X)^{-1}X'(u - \hat{u})$, we have

$$\begin{aligned} V(\beta | \sigma^2, D) &= E\left((\beta - \hat{\beta})(\beta - \hat{\beta})' | \sigma^2, D\right) \\ &= (X'X)^{-1}X'E\left((u - \hat{u})(u - \hat{u})' | \sigma^2, D\right)X(X'X)^{-1} \\ &= \sigma^2(X'X)^{-1} \end{aligned} \quad (7)$$

as the post-data covariance matrix for β given σ^2 . Then using the above definition of σ^2 we have

$$\begin{aligned} E(\sigma^2 | D) &= \frac{1}{n} \left[\hat{u}'\hat{u} + E\left((\beta - \hat{\beta})'X'X(\beta - \hat{\beta}) | D\right) \right] \\ &= \frac{1}{n} (\hat{u}'\hat{u} + kE(\sigma^2 | D)) \end{aligned} \quad (8)$$

or

$$E(\sigma^2 | D) = \frac{\hat{u}'\hat{u}}{n - k} \equiv s^2. \quad (9)$$

Employing (9) in conjunction with (7), we have

$$V(\beta | D) = s^2(X'X)^{-1}. \quad (10)$$

In (4) and (10) are the first and second post-data moments of the elements of the regression coefficient vector, β . It is well known that the proper maxent density given first and second moments is a normal density which in the present

instance is:

$$p(\beta | D) \sim N(\hat{\beta}, s^2(X'X)^{-1}) \quad (11)$$

The normal density in (11), denoted by BMOM(N), can be employed to make post-data probability statements about the possible values of the elements of β , e.g. $\Pr(a < \beta_i < b | D)$, where a and b have known values and β_i is the i -th element of β , which is a solution to the inverse problem posed by Bayes.

Note, as pointed out in Zellner (1996) and Tobias and Zellner (1999) that the proper maxent density for β given σ^2 and D is $N(\hat{\beta}, \sigma^2(X'X)^{-1})$ and that the proper maxent density for σ^2 with first moment equal to s^2 is an exponential density, $f(\sigma^2 | D) = 1/s^2 \exp(-\sigma^2/s^2)$, $0 < \sigma^2 < \infty$, and the marginal density for the regression coefficient vector, denoted by BMOM(1), can be computed. Also, as shown in Tobias and Zellner (1999) and Zellner (1997b), higher order post-data moments of σ^2 can be evaluated and used as moment side conditions in deriving maxent densities. For example, when the first and second moments of the variance parameter are used as side conditions, the proper maxent density is in the truncated normal form, denoted by BMOM(2). Also, moments of functions of the variance parameter, say $\log(\sigma^2)$ and $1/\sigma^2$ can be employed as side conditions to yield an inverted gamma maxent post-data density; see Tobias and Zellner (1999) for details. Then by multiplying the conditional maxent density for β given σ^2 and D by a marginal density for σ^2 the joint density is obtained that can be integrated analytically or numerically to provide various marginal densities for regression coefficients and functions of them; see Tobias and Zellner (1999) for computed examples and use of procedures for selecting among alternative BMOM models

that will be explained below after it is shown how BMOM predictive densities are derived.

2.2 BMOM predictive densities

To derive the post-data predictive moments and density of an $m \times 1$ vector of as yet unobserved future observations, $y_f = X_f \beta + u_f$, where X_f is a given, known $m \times k$ matrix and u_f is an as yet unobserved $m \times 1$ vector of future error terms with unknown values, we make the following assumptions:

Assumption 3 $E(u_f | D') = 0$

Assumption 4 $E(u_f u_f' | \sigma^2, D') = I_m \sigma^2$

Assumption 5 *The elements of u_f are uncorrelated with the elements of β given D'*

where $D' = (y, X, X_f)$. Using the above three assumptions, we have, as pointed out in Zellner (1996),

$$\begin{aligned} E(y_f | D') &= X_f \hat{\beta} \\ V((y_f | \sigma^2, D') &= M \sigma^2 \\ V(y_f | D') &= M s^2 \end{aligned} \tag{12}$$

where $M = I_m + X_f (X' X)^{-1} X_f'$. Using these post-data predictive moments as side conditions, various maxent post-data densities for the future vector, y_f can be derived. For example, using the first and second unconditional moments in (12), the maxent density is $N(X_f \hat{\beta}, M s^2)$. Alternatively, using the mean and conditional covariance matrix in (12), the conditional maxent density is normal with the given mean and conditional covariance matrix. This conditional density can be combined with the various post-data maxent

densities for σ^2 , discussed above. On integrating these joint densities with respect to σ^2 , various marginal densities for the future observations are obtained. An observed value of the vector y_f can be utilized to evaluate alternative predictive densities that can then be employed to form Bayes' factors to appraise alternative BMOM and traditional Bayesian models, as illustrated in Tobias and Zellner (1999). Such Bayes' factors will be used below to help decide the number of terms to retain in a semiparametric series expansion and/or to choose among alternative series expansions.

3. Model selection procedures

Orthonormal series expansions to represent an underlying regression function are known to converge in a mean squared error sense to the unknown regression function as the number of terms in an expansion and the sample size approach infinity. However, in finite samples with a given number of terms in an expansion, there is a need to appraise the adequacy of alternative forms, say different series expansions containing different numbers of terms of measurable variables expressed with and without use of logarithmic or other transformations.

Below, adequacy is evaluated in terms of the values of post-data expectations of estimation and/or predictive loss functions and post-data odds relating to alternative models. As will be seen, expectations of a predictive loss criterion provide both absolute and relative measures of performance, say sample mean squared errors that can be employed to appraise and compare alternative models. The latter capability is also provided by use of post-data odds relating to alternative semiparametric forms. These model selection criteria are implemented below using BMOM analysis and, as is well known,

can also be implemented in a Bayesian approach. Herein, we present three different model selection approaches, namely estimation loss function, predictive loss function and post-data odds approaches. Given the focus of interest, one or another of the approaches will be most appropriate. Also, results of the different approaches will be compared below.

The problems considered include: (a) How many terms to include in a particular series expansion, say a Legendre series? (b) Which series expansion is to be used in representing a regression? And, of course, various combinations of (a) and (b) can be entertained.

We consider the following regression relation for the given data

$$y = X_1\beta_1 + X_2\beta_2 + u \quad (13)$$

where y is an $n \times 1$ vector of observations on a dependent variable, X_1 and X_2 are given $n \times k_1$ and $n \times k_2$ matrices, respectively, with $[X_1 \ X_2]$ of full column rank, u is an $n \times 1$ vector of realized errors and β_1 and β_2 are $k_1 \times 1$ vectors, respectively of regression parameters with the former containing an intercept parameter. We can utilize (13) in connection with model selection problems. Here we consider the problem of how to determine the number of terms to retain in a particular expansion. The first term on the right side of (13) contains k_1 terms and the second an additional k_2 terms. Here the issue is whether the additional terms are needed or, equivalently, if $\beta_2 = 0$.

To analyze the problem of determining whether or not $\beta_2 = 0$ in (13), it is convenient to reparameterize the model as follows

$$y = X_1\phi + \hat{V}\beta_2 + u \quad (14)$$

where $\hat{V} = X_2 - X_1\hat{\Pi}$, with $\hat{\Pi} = (X_1'X_1)^{-1}X_1'X_2$ and $\phi = \beta_1 + \hat{\Pi}\beta_2$. Note that by construction, $X_1'\hat{V} = 0$. Further, using the BMOM regression

approach applied to (14), we have $E(\beta_2 | D) = (\hat{V}'\hat{V})^{-1}\hat{V}'y$ and $V(\beta_2 | D) = s^2(\hat{V}'\hat{V})^{-1}$, where $s^2 = \hat{u}'\hat{u}/(n - k_1 - k_2)$ and $\hat{u}$ the least squares residual vector.

Now if our interest centres on the value of β_2 , namely is it equal to zero or not, given (13) we can consider use of the following quadratic loss function,

$$L(\beta_2, \tilde{\beta}_2) = (\beta_2 - \tilde{\beta}_2)' \hat{V}'\hat{V}(\beta_2 - \tilde{\beta}_2), \quad (15)$$

where $\tilde{\beta}_2$ is some estimate of β_2 . Now we evaluate expected loss given that $\tilde{\beta}_2 = 0$ to obtain,

$$EL(\beta_2, 0) = E\left(\beta_2' \hat{V}'\hat{V}\beta_2 | D\right) = k_2 s^2 + \hat{\beta}_2' \hat{V}'\hat{V}\hat{\beta}_2 \quad (16)$$

With $\tilde{\beta}_2 = \hat{\beta}_2$, the post-data mean of β_2 (an optimal point estimate), expected loss is given by

$$\begin{aligned} EL(\beta_2, \hat{\beta}_2) &= E\left[\left(\beta_2 - \hat{\beta}_2\right)' \hat{V}'\hat{V}\left(\beta_2 - \hat{\beta}_2\right) | D\right] + L_\alpha \\ &= (k_2 s^2 + L_\alpha) \end{aligned} \quad (17)$$

where in evaluating the expectation, $\beta_2 \sim N\left(\hat{\beta}_2, s^2(\hat{V}'\hat{V})^{-1}\right)$ has been used and L_α is the added loss associated with adopting the more complicated model to be assigned by the investigator. Thus, the ratio of expected losses, denoted by C_{12} is just

$$C_{12} = \frac{k_2 s^2 + \hat{\beta}_2' \hat{V}'\hat{V}\hat{\beta}_2}{k_2 s^2 + L_\alpha} = \frac{1 + F}{(1 + Z)} \quad (18)$$

where $z \equiv L_\alpha/k_2 s^2$ and $F = \hat{\beta}_2' \hat{V}'\hat{V}\hat{\beta}_2/k_2 s^2$ is the usual test statistic that appears in a sampling theory test based on normal sampling assumptions and also in certain Bayesian posterior odds expressions relating to the two hypotheses under consideration; see e.g. Kass and Raftery (1995) and Zellner and Siow (1980).

From (18), expected losses are the same when $C_{12} = 1$, or $F = z$ which may be called the indifference value of F . If F is larger than the indifference value, using $\tilde{\beta}_2 = \hat{\beta}_2$ is the act that leads to lower expected loss than using $\tilde{\beta}_2 = 0$. Similarly, if F is smaller in value than the indifference value, then the use of $\tilde{\beta}_2 = 0$ is favoured. Note that a prior view that more complicated models are less probable than simpler models, as implied by the Jeffreys-Wrinch Simplicity Postulate (Jeffreys, 1998), leads us to introduce the penalty term L_α . Such an assignment is also suggested by Lehmann's (1959, p. 52) remark about choosing a significance level in a sampling theory framework, namely that "If one firmly believes the [null] hypothesis to be true, extremely convincing evidence will be required before one is willing to give up this belief, and the significance level will accordingly be set very low." And, of course, the added loss, L_α can take account of whatever relevant information is available and judged suitable to include in the analysis.

In addition, we can use prior probabilities P_1 and P_2 on the hypotheses $H_1 : \beta_2 = 0$ and $H_2 : \beta_2 \neq 0$ to obtain an optimal point estimate of β_2 by solving the following minimization problem:

$$\min_{\tilde{\beta}_2} E \left[L(\beta_2, \tilde{\beta}_2) | D \right] = \min_{\tilde{\beta}} \left(P_1 L(0, \tilde{\beta}_2) + P_2 EL \left((\beta_2, \tilde{\beta}_2) | D \right) \right). \quad (19)$$

If we use a quadratic loss function, $L(\beta_2, \tilde{\beta}_2) = (\beta_2 - \tilde{\beta}_2)' A (\beta_2 - \tilde{\beta}_2)$, where A is a given pds matrix and the normal post-data density for β_2 , $N(\hat{\beta}_2, s^2(\hat{V}'\hat{V})^{-1})$ is employed to evaluate expectations, the optimal estimate is $\tilde{\beta}_2^* = P_2 \hat{\beta}_2 / (P_1 + P_2)$, which is in the form of the Bayes-Stein shrinkage estimate. Note that the larger the odds, P_1/P_2 in favour of $\beta_2 = 0$, the more $\tilde{\beta}_2^*$ is shrunk toward zero.

In the predictive loss function approach we introduce a predictive loss function and calculate its expected value using alternative predictive models. Here we shall just consider prediction based on the observed or given values of the independent variables under the two hypotheses, $\beta_2 = 0$ and $\beta_2 \neq 0$. Given the analysis in section II, the BMOM predictive densities are $N(\hat{y}_{1f}, M_1^{-1} s_1^2)$ and $N(\hat{y}_{2f}, M_2^{-1} s_2^2)$ where $\hat{y}_{1f}$ and $\hat{y}_{2f}$ are least squares predictions under the two hypotheses, $M_1 = (I_m + X_1(X_1'X_1)^{-1}X_1')$, $M_2 = (M_1 + \hat{V}(\hat{V}'\hat{V})^{-1}\hat{V}')$ and s_1^2 and s_2^2 are the sums of squared least squares residuals divided by degrees of freedom, $n - k_1$ and $n - k_1 - k_2$, respectively. We shall employ the following simple loss function, $L(y_f, \hat{y}_{if}) = (y_f - \hat{y}_{if})'(y_f - \hat{y}_{if})$, $i = 1, 2$ where $\hat{y}_{if}$ is the predictive mean, an optimal point prediction relative to quadratic loss functions in general, for each model or hypothesis considered. Thus using $\tilde{\beta}_2 = 0$, expected predictive loss is

$$\begin{aligned} E(L | \tilde{\beta}_2 = 0, D) &= E \left[(y_f - \hat{y}_{1f})'(y_f - \hat{y}_{1f}) | \tilde{\beta}_2 = 0, D \right] \\ &= \text{tr} E \left[(y_f - \hat{y}_{1f})(y_f - \hat{y}_{1f})' | \tilde{\beta}_2 = 0, D \right] \\ &= (n + k_1)s_1^2 \end{aligned} \quad (20)$$

while expected loss under $\tilde{\beta}_2 = \hat{\beta}_2 \neq 0$ is

$$\begin{aligned} E(L | \tilde{\beta}_2 = \hat{\beta}_2, D) &= E \left[(y_f - \hat{y}_{2f})'(y_f - \hat{y}_{2f}) | \tilde{\beta}_2 = \hat{\beta}_2, D \right] + L_\alpha \\ &= \text{tr} E \left[(y_f - \hat{y}_{2f})(y_f - \hat{y}_{2f})' | \tilde{\beta}_2 = \hat{\beta}_2, D \right] + L_\alpha \\ &= (n + k_1 + k_2)s_2^2 + L_\alpha \end{aligned} \quad (21)$$

The ratio of expected losses given in (20) and (21), denoted by R_{12} , is

$$R_{12} = \frac{s_1^2}{s_2^2} \left(\frac{n + k_1}{n + k_1 + k_2 + z} \right) \quad (22)$$

where $z \equiv L_\alpha/s_2^2$. It is seen that with the use of the predictive loss function, the larger is k_2 , all else constant, the less favoured the choice $\tilde{\beta}_2 \neq 0$. That

is, there is a penalty associated with choosing $\tilde{\beta}_2 \neq 0$ that depends on the number of added parameters, k_2 . If R_{12} in (22) satisfies $R_{12} < 1$, $\tilde{\beta}_2 = 0$ is favoured while if $R_{12} > 1$, $\tilde{\beta}_2 \neq 0$ is favoured. Of course, ratios of expected loss can also be computed in a Bayesian framework using relevant predictive densities.

Above in the estimation and predictive loss frameworks, we have employed particular loss functions and in the predictive loss case, prediction given independent variables' values equal to the sample values. Of course, other loss functions and values of the independent variables can also be utilized. Further, more than two variants of the model can be compared in terms of their associated expected losses. These analyses can be augmented by the posterior odds approach that we now take up.

Given that we have the BMOM predictive densities associated with two variants of a model, as above, denoted by $f_i(y_f | H_i, D)$, $i = 1, 2$ a new $m \times 1$ set of observations on y_f denoted by y_f^0 , we can form the usual odds expression:

$$K_{12} = \frac{P_1 f_1(y_f^0 | H_1, D)}{P_2 f_2(y_f^0 | H_2, D)}, \quad (23)$$

that is K_{12} equals the prior odds times the Bayes factor. For the two hypotheses considered above, namely $H_1 : \beta_2 = 0$ and $H_2 : \beta_2 \neq 0$, the Bayes factor (BF) based on the above normal BMOM predictive density is given by

$$BF = \left(\frac{s_2^2}{s_1^2} \right)^{m/2} \frac{|M_2|^{1/2}}{|M_1|^{1/2}} \exp \left(-\frac{SS_1 - SS_2}{2} \right), \quad (24)$$

where $M_1 = (I_m + X_1(X_1'X_1)^{-1}X_1')$, $M_2 = (M_1 + \hat{V}(\hat{V}'\hat{V})^{-1}\hat{V}')$ and $SS_i = (y_f^0 - \hat{y}_i)'M_i(y_f^0 - \hat{y}_i)/s_i^2$, $i = 1, 2$. The expression in (24) involves quantities appearing in the ratio of expected predictive losses in (22) and in addition, the quantities SS_i , a weighted sum of squared prediction errors that

is similar to empirical MSEs that many use in appraising alternative prediction procedures in practice.

Since the posterior odds in (23) is equal to the ratio of post-data probabilities associated with the two hypotheses, as is well known, with a 2×2 loss structure, these probabilities can be employed to minimize expected loss in choosing between the two models. See, e.g. DeGroot (1970), Berger (1985) and Zellner (1971). Again, more than two models can also be compared using posterior odds.

Last, as Good (1950) and Kullback (1959) point out, the average log odds is related to cross entropy by the following relation

$$W_{12} = \int \log K_{12} f_1(y_f | D') dy_f - \log \frac{\pi_1}{\pi_2} = \int f_1(y_f | D') \log \frac{f_1(y_f | D')}{f_2(y_f | D')} dy_f. \quad (25)$$

It is seen that the average log prior odds minus the prior odds is just the cross entropy of model 1 relative to model 2. Further, a symmetric cross entropy expression is given by

$$W = W_{12} + W_{21} = CE(f_1, f_2) + CE(f_2, f_1), \quad (26)$$

which is the Jeffreys-Kullback-Leibler distance measure. Since these average log odds minus the log prior odds, or the “weight of the evidence” are interesting measures relating to alternative models, they have been computed in Tobias and Zellner (1999).

Above we have utilized particular post-data BMOM densities to illustrate various model selection techniques. Of course, other densities reflecting different moment constraints can be employed. See section IV for computed examples. In addition, BMOM models are also compared to Bayesian models based on assumed forms for likelihood functions and particular prior densities.

We now turn to the problem of comparing and choosing between or among alternative semiparametric series expansions. The non-nested models to be compared are

$$y = W_1\beta_1 + u_1 \quad (27)$$

and

$$y = W_2\beta_2 + u_2. \quad (28)$$

As above, we can utilize a predictive loss function approach in making a selection between two alternative expansions in (27) and (28) or a BMOM post-data odds approach given that we have new data or split the given sample of data.

We shall employ the loss function $(y_f - \hat{y}_{if})'(y_f - \hat{y}_{if})/m$, a mean squared predictive error loss function, where $\hat{y}_{if}$ is the mean under model i of the predictive density for an $m \times 1$ vector of future observations, y_f . In particular, $\hat{y}_{if} = W_{if}\hat{\beta}_i$, $i = 1, 2$ for the relations in (27) and (28) respectively. Then expected losses for models H_i , $i = 1, 2$ are given by

$$\begin{aligned} E(L | H_i, D) &= E[(y_f - \hat{y}_{if})'(y_f - \hat{y}_{if})/m | H_i, D] \\ &= \text{tr} (I_m + W_{if}(W_i'W_i)^{-1}W_{if}') s_i^2/m \\ &= (m + a_i) s_i^2/m \end{aligned} \quad (29)$$

where $a_i = \text{tr} (W_{if}(W_i'W_i)^{-1}W_{if}')$ and $s_i^2 = (y_f - W_i\hat{\beta}_i)'(y_f - W_i\hat{\beta}_i)/(n - k_i)$. Thus, the ratio of expected losses, R_{12} is

$$R_{12} = \frac{s_1^2}{s_2^2} \left(\frac{m + a_1}{m + a_2} \right). \quad (30)$$

The value of R_{12} depends directly on $s_i^2(m + a_i)$, the sum of predictive variances of the elements of the $m \times 1$ vector of future observations y_f for the model H_i , $i = 1, 2$. If $R_{12} < 1$, H_1 is favoured, while if $R_{12} > 1$, H_2

is favoured. Of course, predictive loss functions other than the one used above can be employed and losses to reflect penalties for complexity can be added.

Given that we have the predictive density functions for a vector of future observations for the models in (27) and (28) and have observed values of the elements of y_f and W_{if} , $i = 1, 2$, we can compute the post-data odds,

$$K_{12} = \frac{P_1 f_1(y_f^0 | H_1, D)}{P_2 f_2(y_f^0 | H_2, D)} \quad (31)$$

where y_f^0 is the observed value of y_f and P_1/P_2 is the prior odds for models M_1 and M_2 in (27) and (28) respectively. As a special important case, if f_1 and f_2 in (30) are BMOM normal predictive densities, $N(\hat{y}_{if}, V_i)$, $i = 1, 2$ $\log(K_{12})$ is given by

$$\begin{aligned} \log K_{12} = & \log(P_1/P_2) - (1/2) \log \frac{|V_1|}{|V_2|} - (1/2)(y_f^0 - \hat{y}_{1f})' V_1^{-1} (y_f^0 - \hat{y}_{1f}) \\ & + (1/2)(y_f^0 - \hat{y}_{2f})' V_2^{-1} (y_f^0 - \hat{y}_{2f}) \end{aligned} \quad (32)$$

If we write $K_{12} = \pi_1/\pi_2$ where π_i is the probability associated with model H_i , these probabilities can be employed to obtain minimum expected predictive loss point predictions relative to two models, say (27) and (28), that are averages of the individual model's optimal point predictions. That is, if we have a quadratic predictive loss function, $L(y_f, \tilde{y}_f) = (y_f - \tilde{y}_f)' Q (y_f - \tilde{y}_f)$, where Q is a pds matrix, we have

$$\begin{aligned} E(L) &= \pi_1 E(L | H_1) + \pi_2 E(L | H_2) \\ &= \pi_1 E[(y_f - \tilde{y}_f)' Q (y_f - \tilde{y}_f) | H_1] + \pi_2 E[(y_f - \tilde{y}_f)' Q (y_f - \tilde{y}_f) | H_2] \\ &= \sum_{i=1}^2 \pi_i [(\tilde{y}_f - \hat{y}_{if})' Q (\tilde{y}_f - \hat{y}_{if}) + \text{tr}(Q V_i)] \end{aligned} \quad (33)$$

and on minimizing (33) with respect to $\tilde{y}_f$, the optimal point prediction is

$$\tilde{y}_f^* = \frac{\pi_1 \hat{y}_{1f} + \pi_2 \hat{y}_{2f}}{\pi_1 + \pi_2} = \frac{K_{12} \hat{y}_{1f} + \hat{y}_{2f}}{K_{12} + 1} \quad (34)$$

where $\hat{y}_{if}$ is the predictive mean for model H_i . Thus, if the support for H_1 is very low, that is K_{12} is close to zero in value, the optimal point prediction in (34) reduces to that for H_2 , namely $\hat{y}_{2f}$, the mean of its predictive density. On the other hand, if the data support H_1 strongly as reflected by a very large value for the posterior odds K_{12} , then (34) reduces to $\hat{y}_{1f}$, the optimal point prediction for H_1 , its predictive mean. This procedure permits an averaging over more than two models and/or over BMOM and Bayesian models' predictions.

4. Computed examples

In this section, we perform some Monte-Carlo experiments by generating data from a CES production function. Capital (K) and Labour (L) are drawn independently and uniformly on the unit interval $[0, 1]$. Given the capital and labour draws, output (y) is then generated according to the equation

$$y_i = \gamma (\delta K_i^{-p} + (1 - \delta)L_i^{-p})^{-v/p} \exp u_i, \quad (35)$$

where the parameter values $\gamma = \exp(1.0564)$, $\delta = .4064$, $p = .6042$, $v = .8222$ are used and the u_i 's are assumed to be normally distributed with zero mean and variance-covariance matrix $.01I_N$. These parameter values are taken from Ryu (1993).

We apply our analysis to two sample sizes. For the first sample size, we use 25 observations to fit the model and an additional 25 observations to evaluate the model's predictive performance. In the second case, we use 1,000 observations to fit the model and 150 observations for prediction. For each sample size, we consider estimation by taking logarithms of the dependent and independent variables. We also make use of Fourier series expansions and the

Fourier Flexible form proposed by Gallant (1981), and thus it is necessary to transform the data to lie in the closed interval $[0, 2\pi]$ for these models.

For nested hypotheses testing, i.e., $H_0 : \beta_2 = 0$, as described in section 3, we compute posterior odds based on prior odds equal to one, report F-statistics for the estimation loss criterion in (18), and compute predictive loss (22). For non-nested testing, i.e., comparing alternative series expansions, we compute posterior odds. As suggested in section 3, we can also use cross entropy and the Kullback-Leibler distance function as a method of model comparison, as done in Tobias and Zellner (1999). Results are presented from both a Bayesian approach based on an improper diffuse prior and iid normal likelihood function, and also using BMOM.

To assess the fit of the model, we compute root mean square error (RMSE), root mean absolute error (RMAE) and R^2 . We define RMSE as $\sqrt{1/N \sum (\hat{y}_i - E(y_i | X_i))^2}$, where $\hat{y}_i$ is the fitted value for the logarithm of output, and $E(y_i | X_i)$ is the logarithm of the CES production function given in (35). We define RMAE similarly, using absolute deviations from the CES conditional mean function.

We also carry along the “true” model in our analysis to obtain absolute performance measures. That is, we assume the parameter values in (35) and the functional form are known, as well as the distribution of the error terms. Then we compute posterior odds as in equation (23) with the true model in the denominator. From equation (35), we see that $\log y$ is normally distributed, and we compare the ordinate of this multivariate normal predictive density with the ordinate of alternative predictive densities to compute posterior odds based on prior odds equal to one.

We consider a variety of alternative series expansions as candidate models and assess the performance of each. the functional forms used are enumerated below

$$\text{Poly. Series } y_i = \sum_{\{(j, m)\} \in \Omega_1} \alpha_{jm} K_i^j L_i^m + \epsilon_{1i}$$

$$\text{Fourier } y_i = \gamma_0 + \sum_{h \in H_1} (\beta_{h1} \cos(h' X_i) + \beta_{h2} \sin(h' X_i)) + \epsilon_{2i}$$

$$\text{Gallant } y_i = \sum_{\{(j, m)\} \in \Omega_1} \alpha_{jm} K_i^j L_i^m + \sum_{h \in H_1} (\beta_{h1} \cos(h' X_i) + \beta_{h2} \sin(h' X_i)) \epsilon_{3i}$$

where

$\Omega_l = \{(j, m) : j + m \leq l, j \geq 0, m \geq 0\}$, $H_2 = \{[1 \ 0]', [0 \ 1]'\}$, $H_3 = \{[1 \ 0]', [0 \ 1]', [1 \ 1]'\}$, and $H_4 = \{[1 \ 0]', [0 \ 1]', [1 \ 1]', [1 \ -1]'\}$. Here, the H_i denote the set of multi-indices used in the Fourier Series and Gallant Forms, Ω_i denotes the degree of our polynomial series expansion, and $X_i = [K_i \ L_i]'$. It is worthwhile to note that the polynomial expansion used here is similar to the Legendre polynomial expansion. By choosing the appropriate linear combination of capital and labour, we can form the terms of the continuous Legendre series expansion and obtain coefficients which are not affected by the number of terms in the expansion following the method in Ryu (1993). In the empirical calculations which follow, u , K , and L are replaced by their natural logarithms.

We see from the tables in the appendix the following "facts" when comparing the alternative models:

- Fourier expansions, which are intended to approximate 2π -periodic functions, do not perform well in this analysis.

- The polynomial series expansion performs the best among the alternative models considered, and the addition of cyclical components to the polynomial series does not improve the models' performance in general.
- Bayesian models are not uniformly favoured relative to BMOM models in tables 3 and 4 even though the data were generated in accord with the assumed normal likelihood function.
- Posterior odds and the predictive loss criterion show that we can "overfit" the regression function in small samples. When $N = 25$, the quadratic polynomial series is most favoured by the data, while cubic expansion is most favoured by the data when $N = 1,000$. Further, with $N = 1,000$, the "true" model is only favoured over the BMOM(1) model by a factor of 2:1.

The good performance of the polynomial series models relative to the true model is not surprising since the linear model gives the Cobb-Douglas form and the quadratic gives a translog approximation, which are first and second-order approximations, respectively to the CES function. As evident from (22), the predictive loss function "penalizes" the addition of extra parameters heavily in smaller samples, while this penalty tends to "wash out" as $n \rightarrow \infty$. When $N = 25$, the predictive loss function supports the hypothesis that the cubic and quadratic terms are zero, but does not support the hypothesis that only the quadratic terms are zero.

5. Summary and conclusions

We have demonstrated how the BMOM approach can be employed to analyze a canonical regression problem without the need for an assumed likelihood function and a prior density for its parameters. When the form of the regression function is unknown, we showed how alternative series expansions can be used in the BMOM approach to approximate the underlying conditional mean function. Operational estimation, prediction and model selection procedures

were developed and applied. Relative expected loss, post-data odds and predictive loss were employed to evaluate the performance of alternative BMOM and Bayesian models and to choose the series expansion that is most favoured by the given data. In analyses of generated data from a known underlying model, it was found that certain series expansions were preferred to others in small to moderate sized samples. Our calculations also showed that as the sample size grows and the order of the polynomial series expansion is increased, the post-data odds of the approximating series expansion to the "true" model approaches 1:1. This result is to be expected given the large sample convergence results for series expansions and the consistency property of post-data odds. Last, we believe that our BMOM techniques will be useful to analysts who are concerned about which series expansion to use, how many terms to retain and whether assumptions about the forms of likelihood functions are appropriate.

As regards future research, the techniques reported in this paper can be developed further to apply to multivariate semiparametric models and to models involving discrete random variables, e.g. logit, probit and other forms of discrete choice models, etc. Second, more work has to be done to extend our methods to apply to transformations of variables, say the well-known Box-Cox transformation and to those appearing in generalized production functions, e.g. $\log y + \theta y$. Indeed, use of an appropriate transformation may do much to help improve the ability of semiparametric forms to approximate more closely underlying functions when sample sizes are not large. Third, with respect to the model selection procedures presented in this paper, it would be desirable to study the finite sample properties, or average "operating characteristics" of these procedures in small to moderate-sized samples. Last, in our computed

examples, error terms were independently drawn from a zero mean, constant variance normal distribution. Further study of the relative properties of BMOM and of “normality-based” diffuse prior traditional Bayesian procedures when data are non-normal would be most valuable.

6. Appendix

Notes on tables 1–4

Tables 1–2 present alternative measures of fit (R^2 , RMSE and RMAE) for the polynomial series, Fourier series, and Gallant’s Fourier Flexible form for generated data sets consisting of 25 and 1,000 observations. RMSE and RMAE are calculated as described in section 4. Tables 3–4 present posterior odds based on equal prior odds (equation 31), F -statistics associated with the estimation loss criterion (equation 18), and predictive loss (equation 22) calculations for both generated data sets. See p. 11 for definitions of BMOM(N), BMOM(1) and BMOM(2). For the calculations in table 3, 25 observations are used to fit the model and 25 observations are used for prediction. Table 4 uses 1,000 observations to fit the model and 150 observations for a hold-out sample. For the Fourier series and Gallant’s Fourier flexible form, the logarithm of capital and labour are scaled to lie in the interval $[0, 2\pi]$. Both the F -statistic and the predictive loss values test the given model against the model with the fewest number of parameters in each class. For example, the value for “Poly. Series (Quadratic)” tests whether the added second-degree polynomial terms are jointly zero, and similarly, the “Poly. Series (Cubic)” tests if the added quadratic and cubic (polynomials of degree 3) terms are zero. The posterior odds give the odds relative to the true model, where the ordinate of the true model is obtained as described in section 4. Since all the odds are less than

one, no model is preferred over the “true” model, although the higher-order polynomial series expansions come close to the true model, particularly when 1,000 observations are used in our experiments. The results in table 3 were also computed for the Fourier and Gallant forms, but are not reported. We found that the Fourier series expansions performed very poorly relative to the polynomial series expansion, and the inclusion of cyclical terms in the Gallant forms were not favoured by the data. For more details on the functional forms used in estimating the CES production function, see section 4.

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Table 1. Measures of fit for alternative models. $N = 25$.
All variables expressed as natural logarithms

Functional form	R^2	RMSE	RMAE
Poly. Series (Lin.) Cobb-Douglas)	.964	.111	.306
Poly. Series (Quadratic)	.987	.054	.209
Poly. Series (Cubic)	.987	.054	.207
Fourier (2)	.375	.506	.601
Fourier (3)	.524	.447	.567
Fourier (4)	.542	.441	.551
Gallant (Lin, 2)	.939	.150	.318
Gallant (Quad, 2)	.984	.088	.264
Gallant (Lin, 3)	.940	.148	.318
Gallant (Quad, 3)	.984	.088	.265

Table 2. Measures of fit for alternative models. $N = 1,000$
All variables expressed as natural logarithms

Functional form	R^2	RMSE	RMAE
Poly. Series (Lin.) (Cobb-Douglas)	.946	.168	.310
Poly. Series (Quadratic)	.986	.028	.140
Poly. Series (Cubic)	.987	.019	.115
Fourier (2)	.294	.682	.658
Fourier (3)	.295	.681	.657
Fourier (4)	.297	.680	.662
Gallant (Lin, 2)	.835	.320	.421
Gallant (Quad, 2)	.911	.226	.325
Gallant (Lin, 3)	.835	.319	.419
Gallant (Quad, 3)	.911	.226	.324

Table 3. Posterior odds, F -statistics and predictive loss for alternative series expansions. $N = 25$, all variables expressed as natural logs

Model	Approach	Posterior odds	F-Stat	Pred. Loss
Poly. Series (Linear) (Cobb-Douglas)	BMOM(N)	1.30e(-11)	Base Model	Base Model
	TB	1.21e(-9)		
	BMOM(1)	2.50e(-9)		
	BMOM(2)	3.72e(-11)		
Poly. Series (Quadratic)	BMOM(N)	9.03e(-3)	10.5	2.08
	TB	1.07e(-2)	10.5	2.06
	BMOM(1)	5.32e(-3)	10.5	2.08
	BMOM(2)	9.56e(-3)	10.5	2.08
Poly. Series (Cubic)	BMOM(N)	1.76e(-4)	3.80	.800
	TB	1.12e(-4)	3.80	.763
	BMOM(1)	4.69e(-5)	3.80	.800
	BMOM(2)	2.35e(-5)	3.80	.800

Table 4. Posterior odds, F -statistics and predictive loss for alternative series expansions. $N = 1,000$, all variables expressed as natural logs

Model	Approach	Posterior odds	F-Stat	Pred. Loss
Poly. Series (Linear) (Cobb-Douglas)	BMOM(N)	8.19e(-39)	Base Model	Base Model
	TB	8.73e(-39)		
	BMOM(1)	2.19e(-39)		
	BMOM(2)	8.09e(-39)		
Poly. Series (Quadratic)	BMOM(N)	3.79e(-5)	951	3.85
	TB	1.06e(-4)	951	3.85
	BMOM(1)	2.85e(-3)	951	3.85
	BMOM(2)	3.82e(-5)	951	3.85
Poly. Series (Cubic)	BMOM(N)	2.24e(-2)	429	3.98
	TB	5.10e(-2)	429	3.98
	BMOM(1)	5.38e(-1)	429	3.98
	BMOM(2)	2.26e(-2)	429	3.98

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